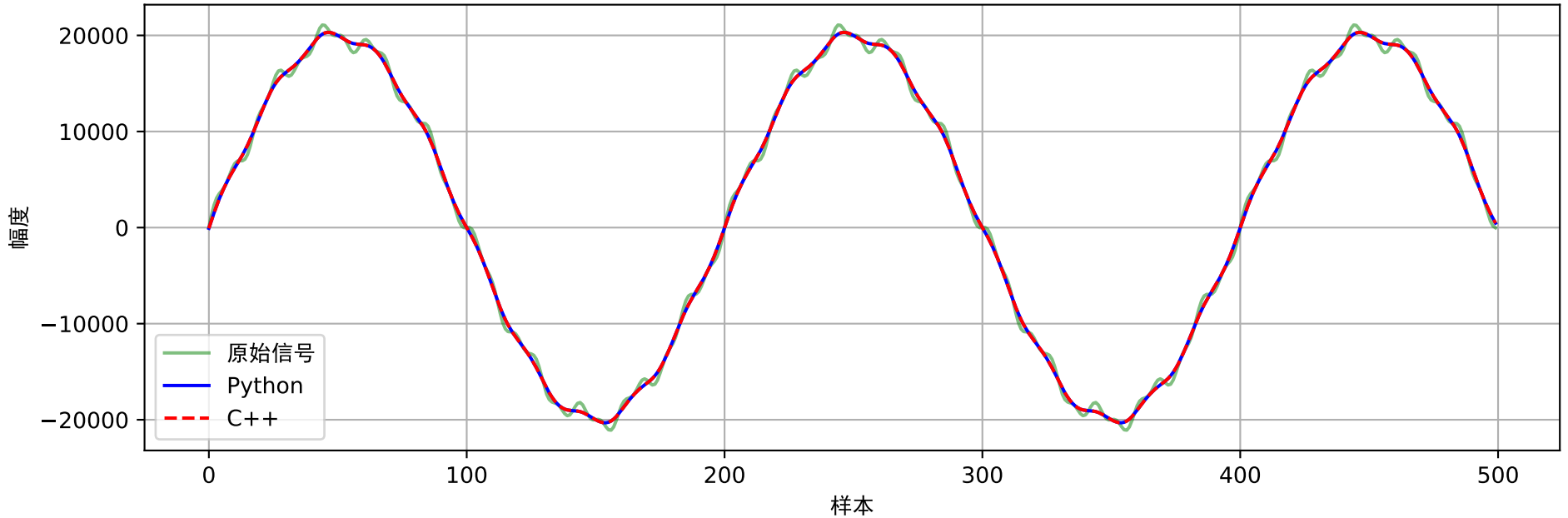
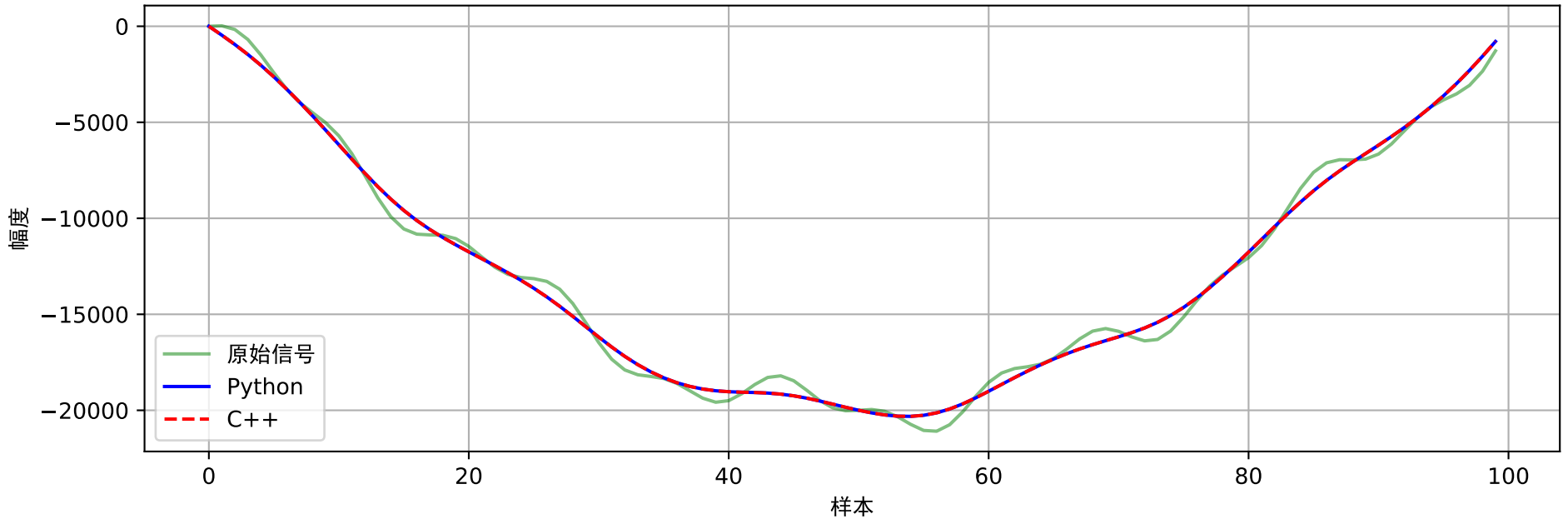


FiltFilt 滤波器实现比较: Python vs C++: 2nd 阶低通滤波器 (Wn=0.1)

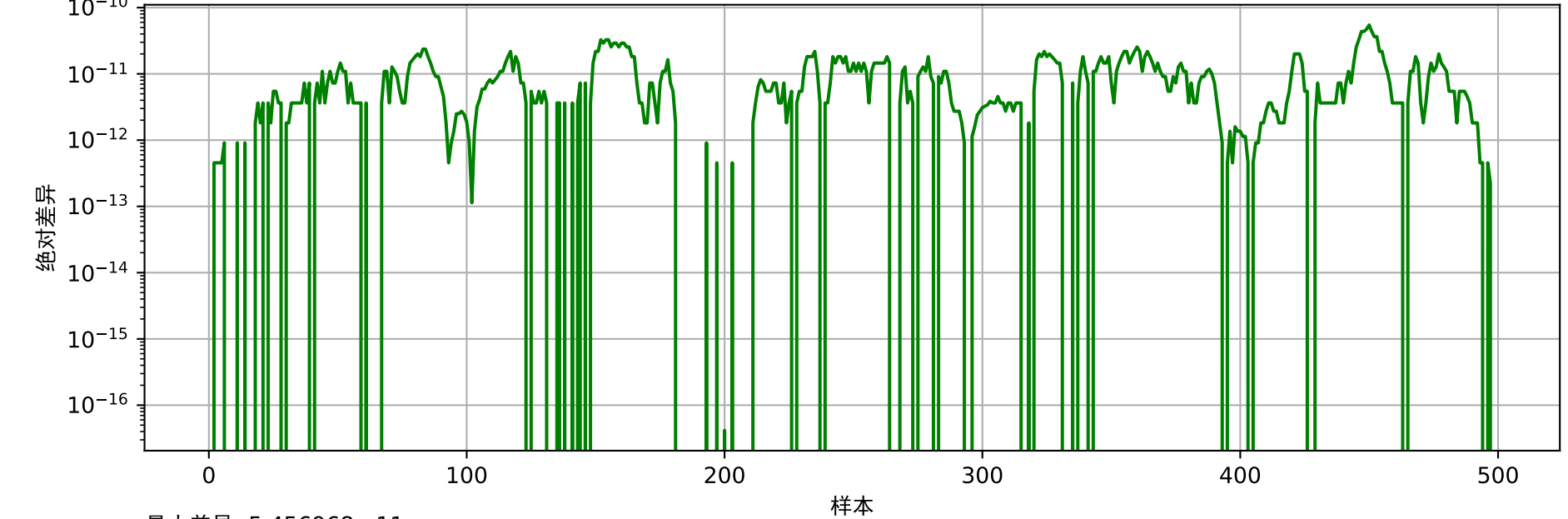
信号对比(前500个样本)



局部放大 (样本 100-200)



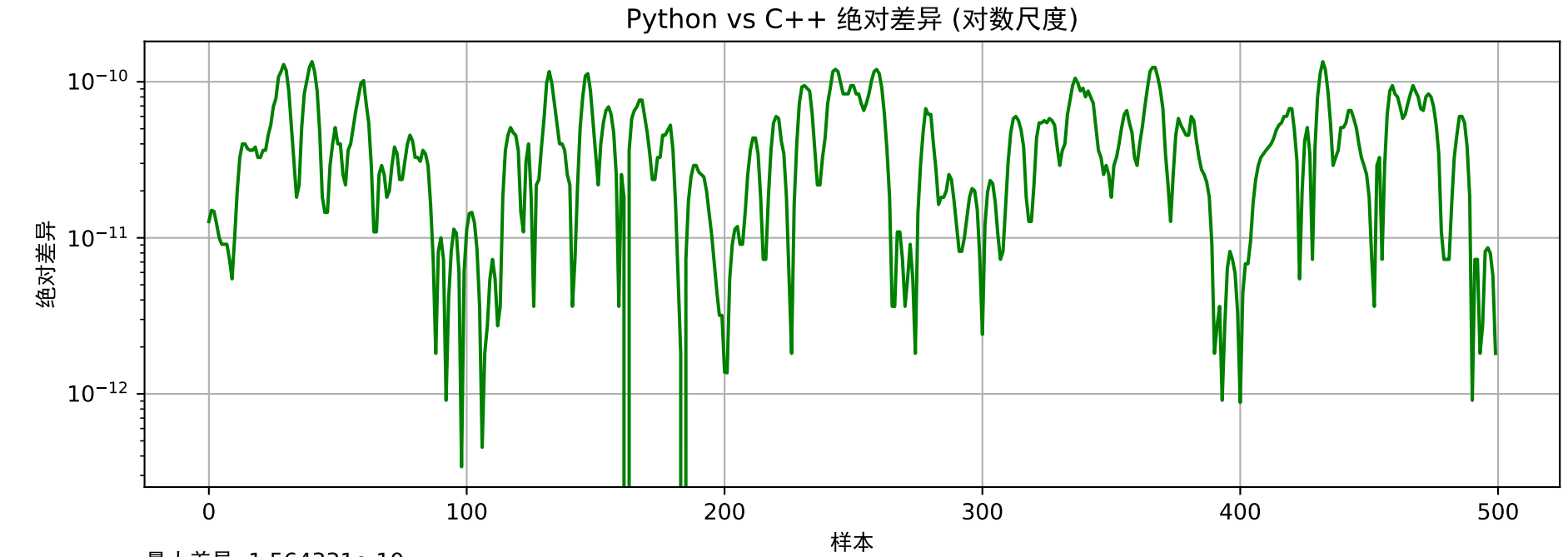
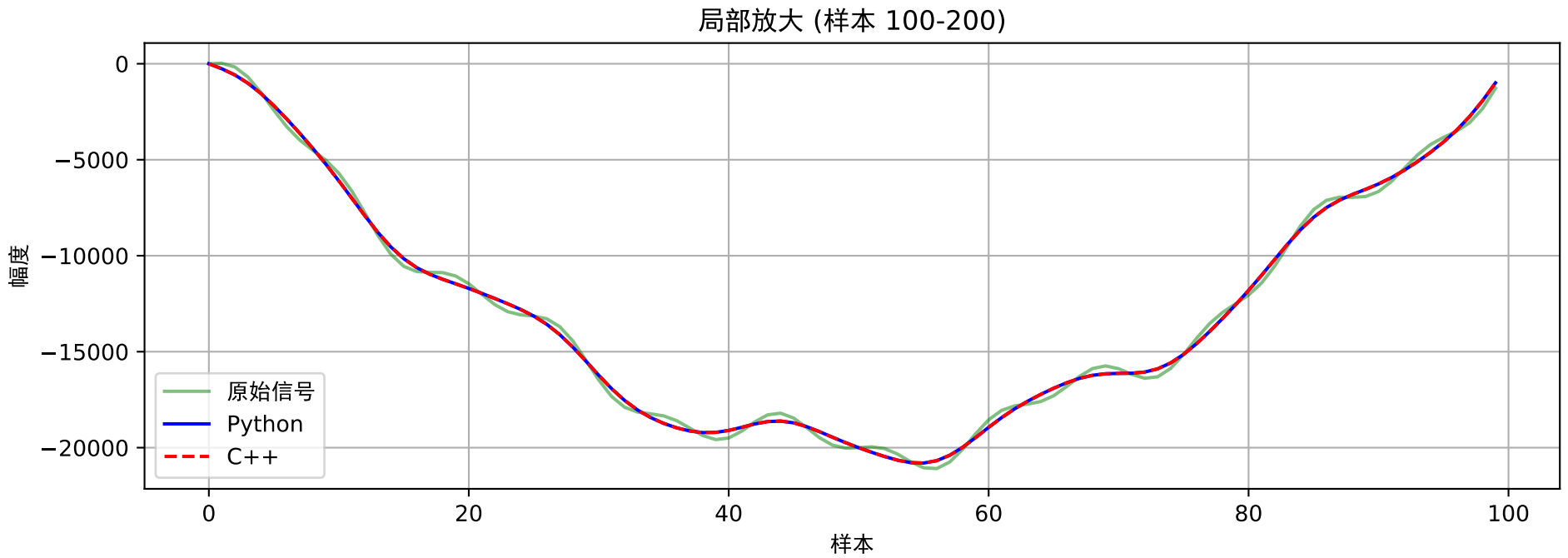
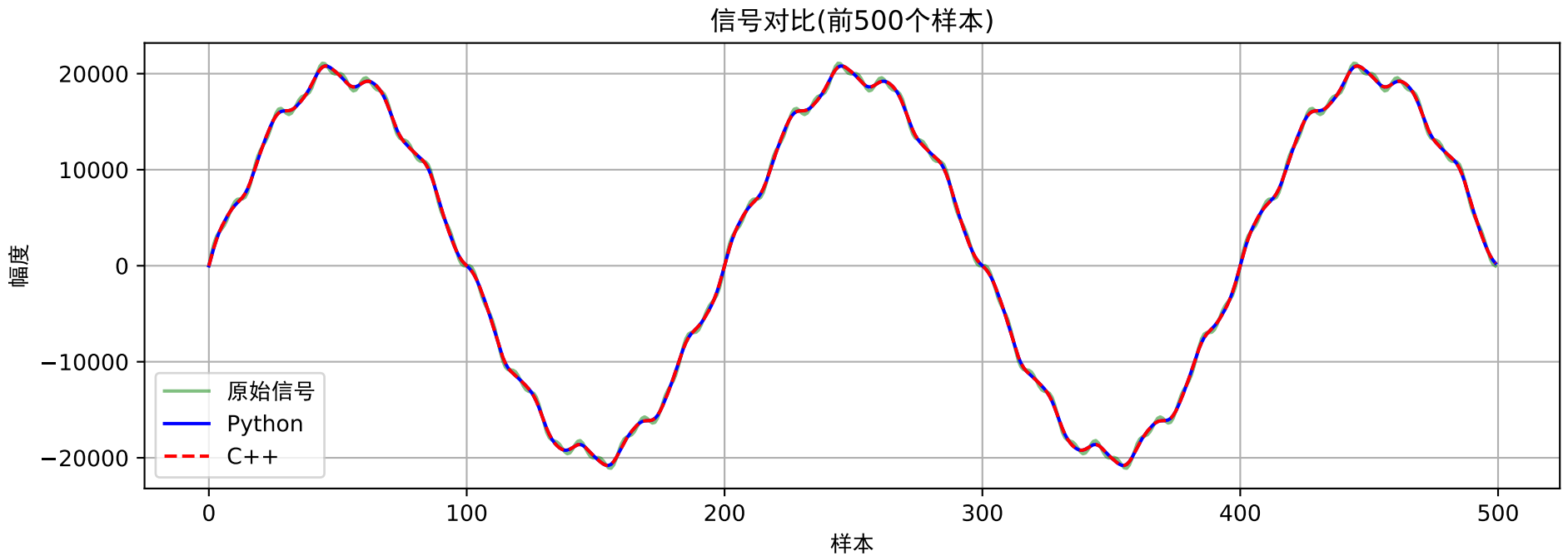
Python vs C++ 绝对差异 (对数尺度)



最大差异: 5.456968e-11  
平均差异: 7.305189e-12  
标准差: 8.364917e-12

结果匹配: ✓

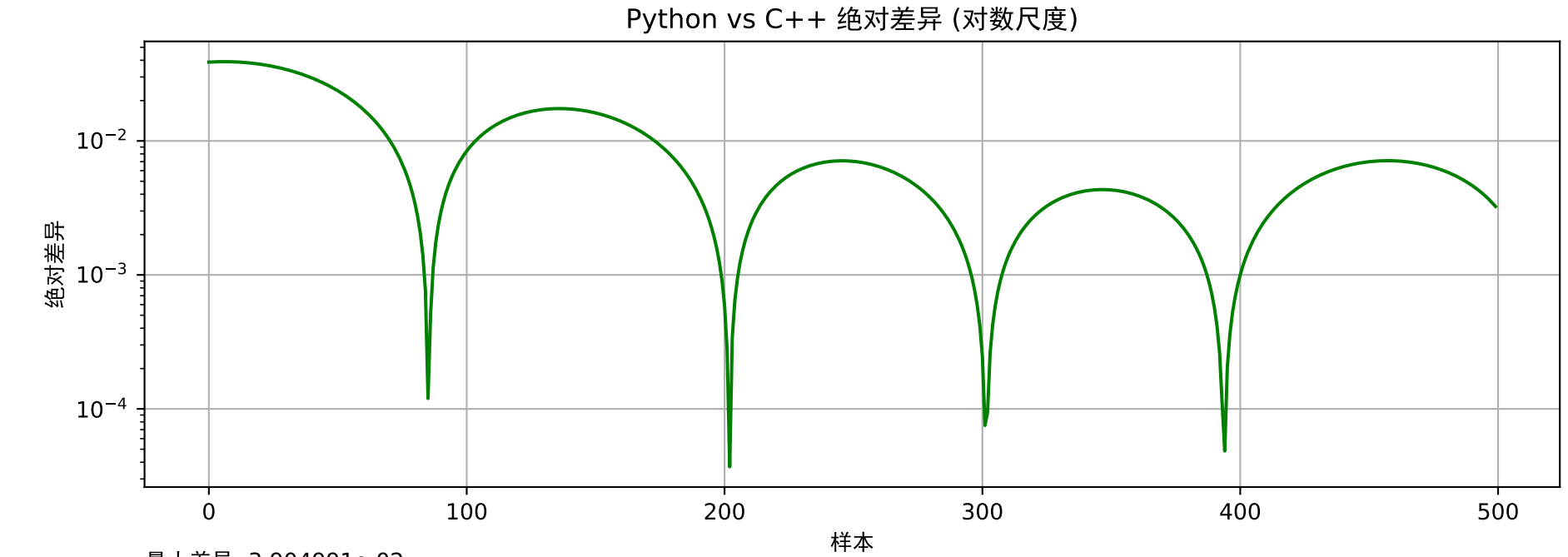
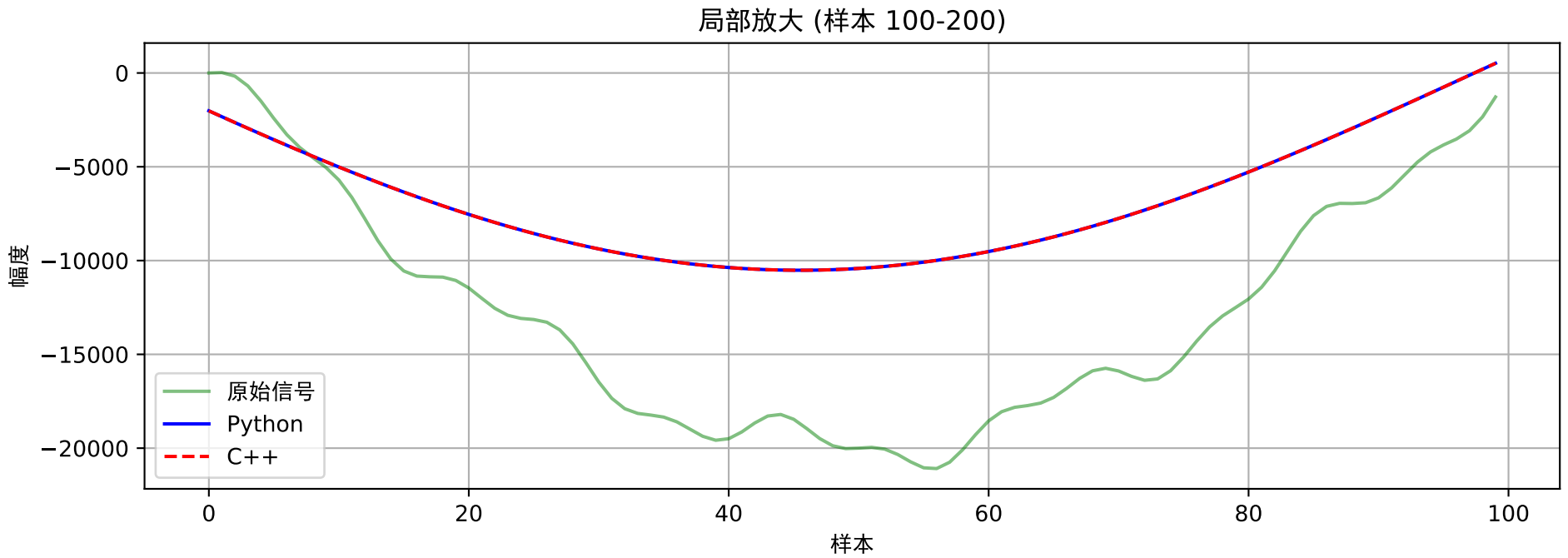
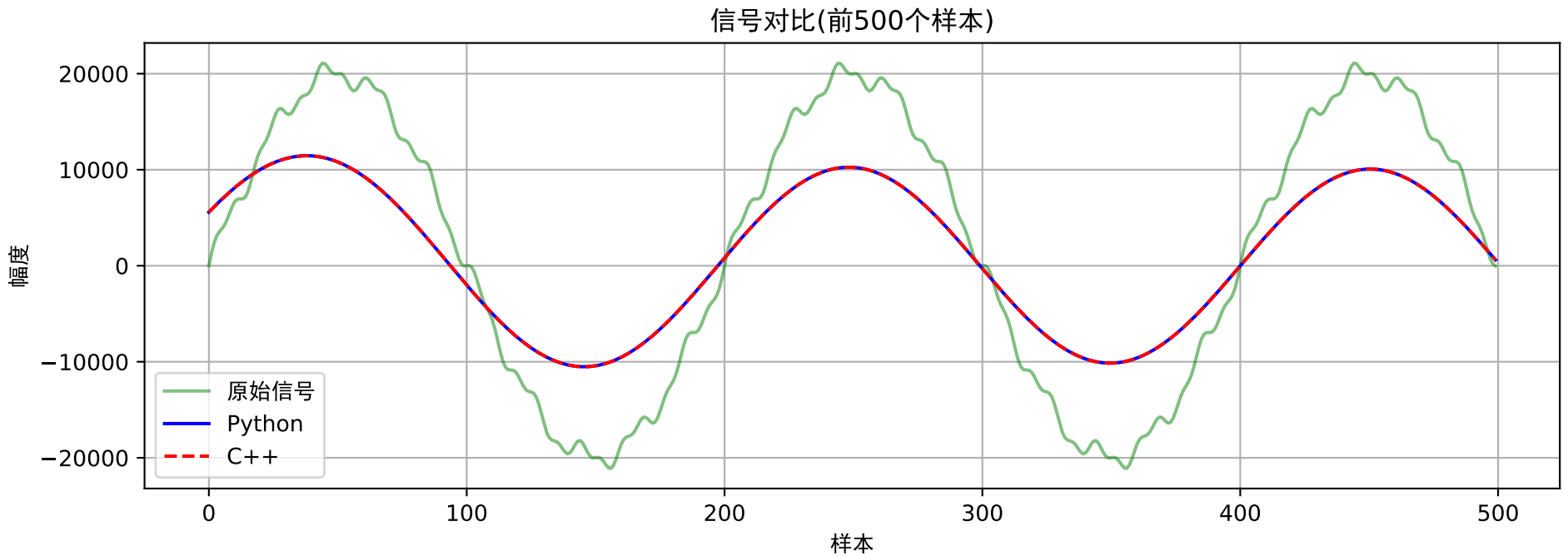
FiltFilt 滤波器实现比较: Python vs C++: 4th 阶低通滤波器 (Wn=0.2)



最大差异: 1.564331e-10  
平均差异: 4.025834e-11  
标准差: 3.278857e-11

结果匹配: ✗

FiltFilt 滤波器实现比较: Python vs C++: 6th 阶低通滤波器 (Wn=0.01)

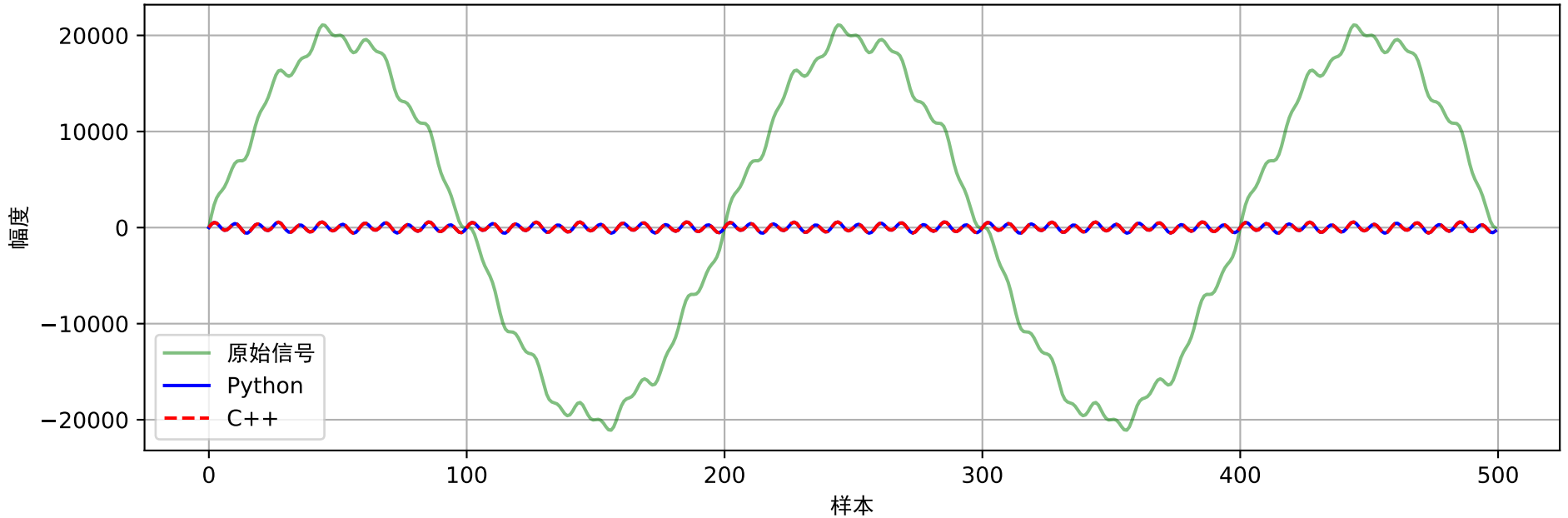


最大差异: 3.904991e-02  
平均差异: 1.084110e-02  
标准差: 1.095905e-02

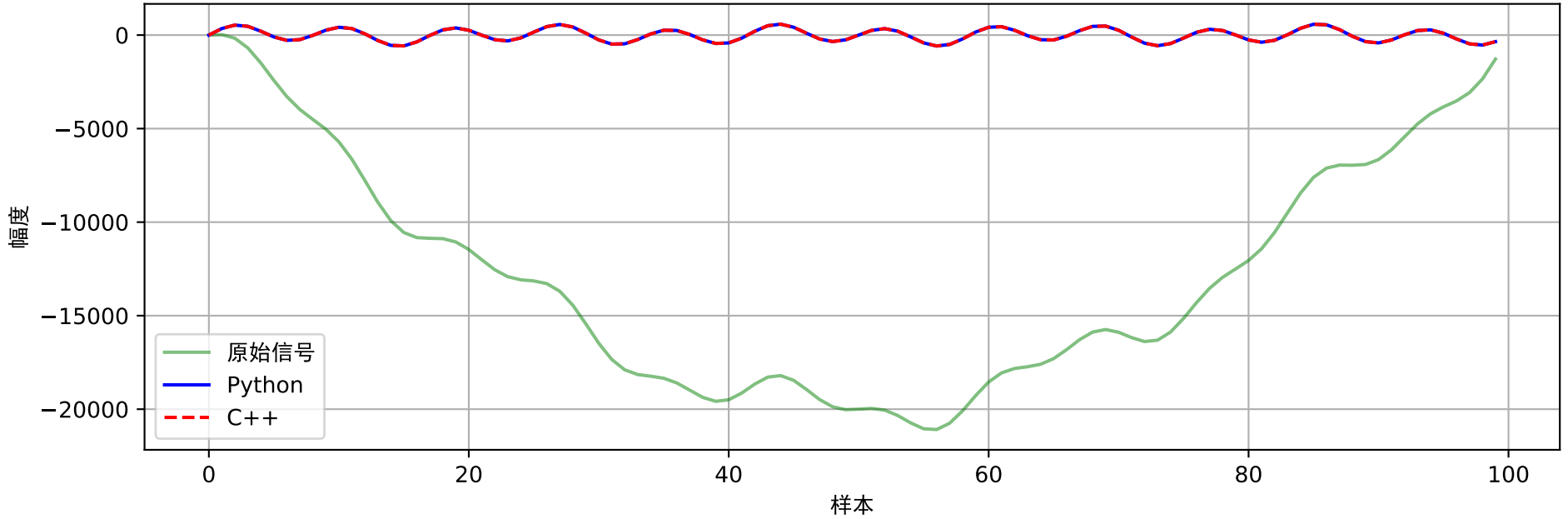
结果匹配: ✗

FiltFilt 滤波器实现比较: Python vs C++: 2nd 阶高通滤波器 ( $W_n=0.15$ )

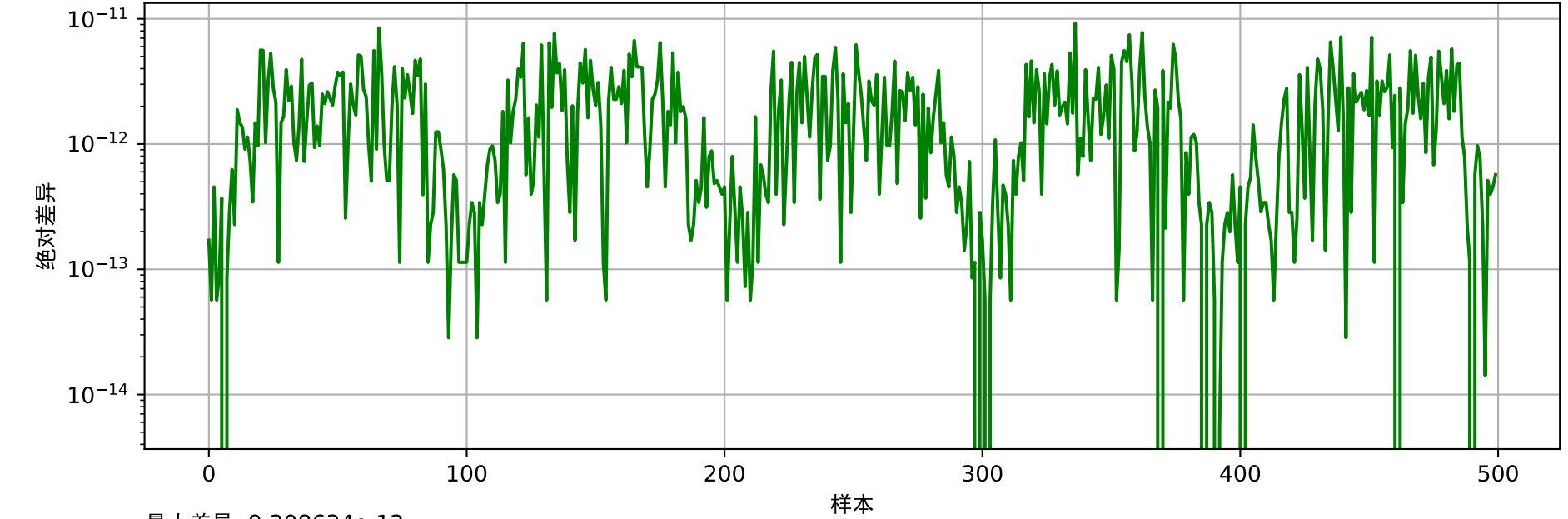
信号对比(前500个样本)



局部放大 (样本 100-200)



Python vs C++ 绝对差异 (对数尺度)

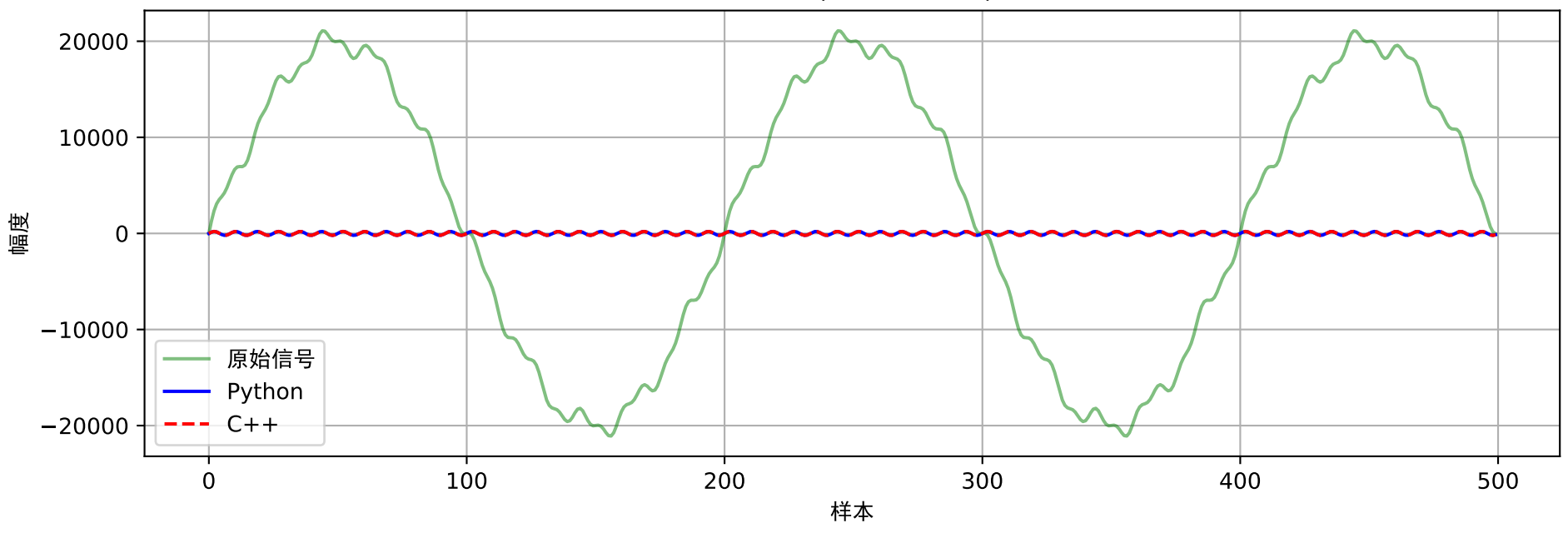


最大差异:  $9.208634e-12$   
平均差异:  $1.852989e-12$   
标准差:  $1.723410e-12$

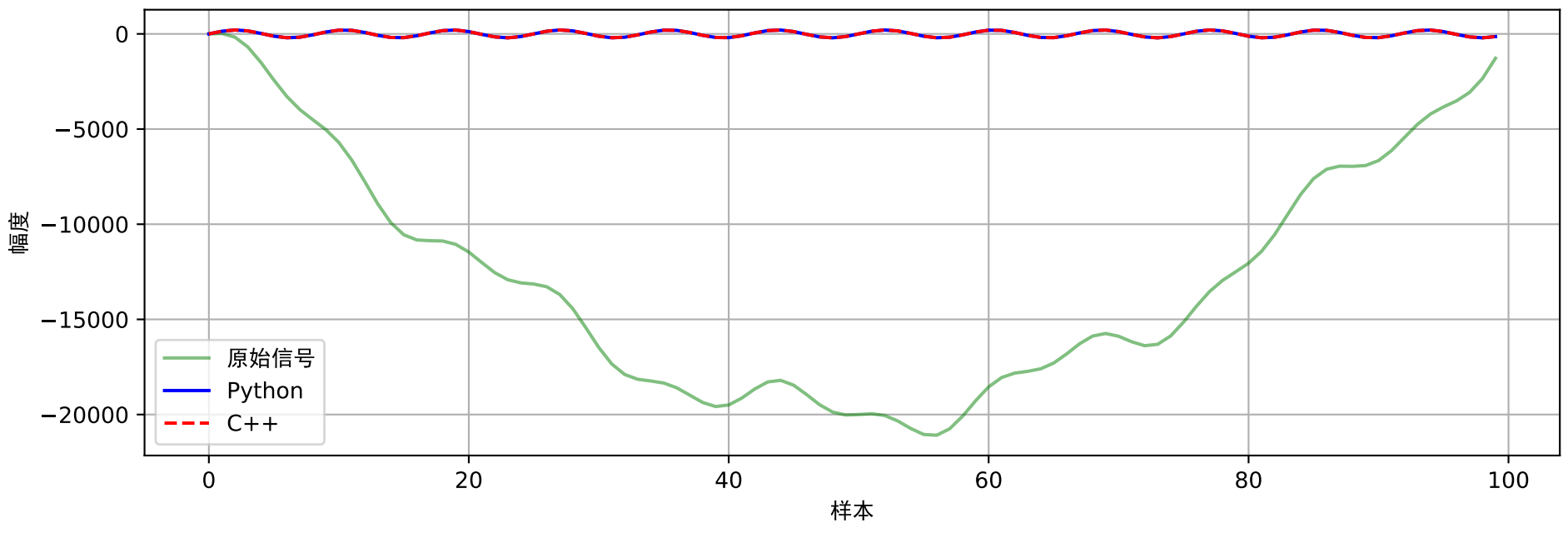
结果匹配: ✓

# FiltFilt 滤波器实现比较: Python vs C++: 4th 阶高通滤波器 (Wn=0.25)

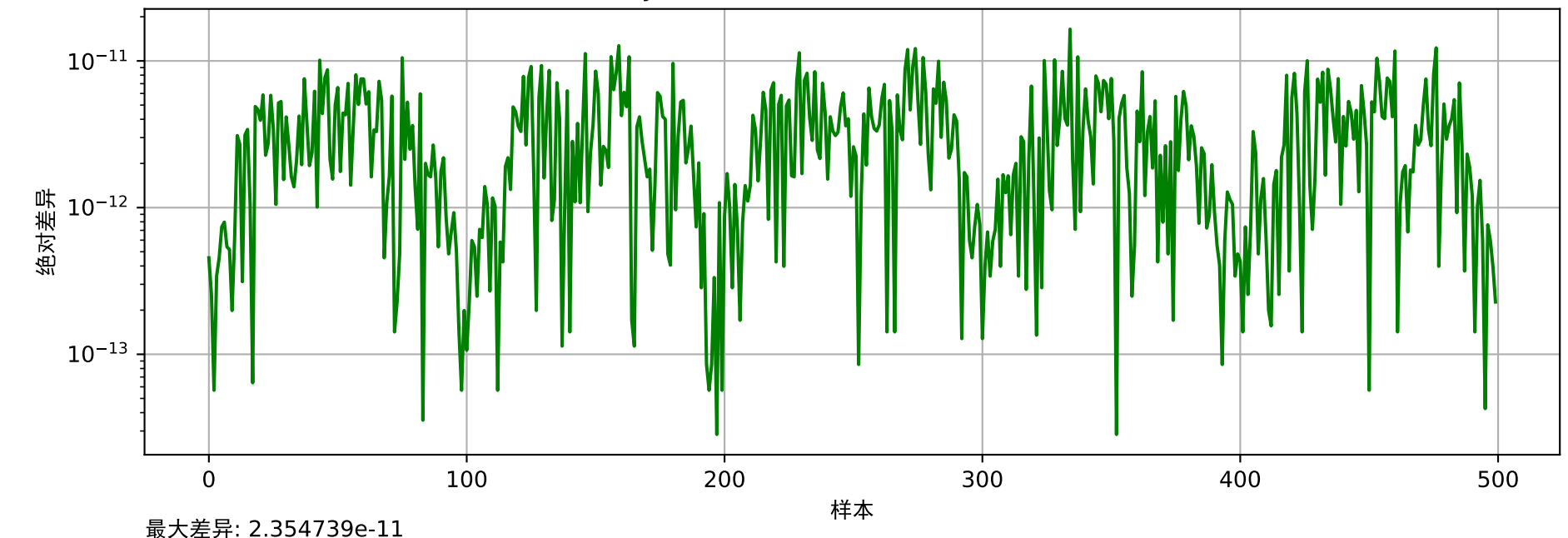
## 信号对比(前500个样本)



## 局部放大 (样本 100-200)



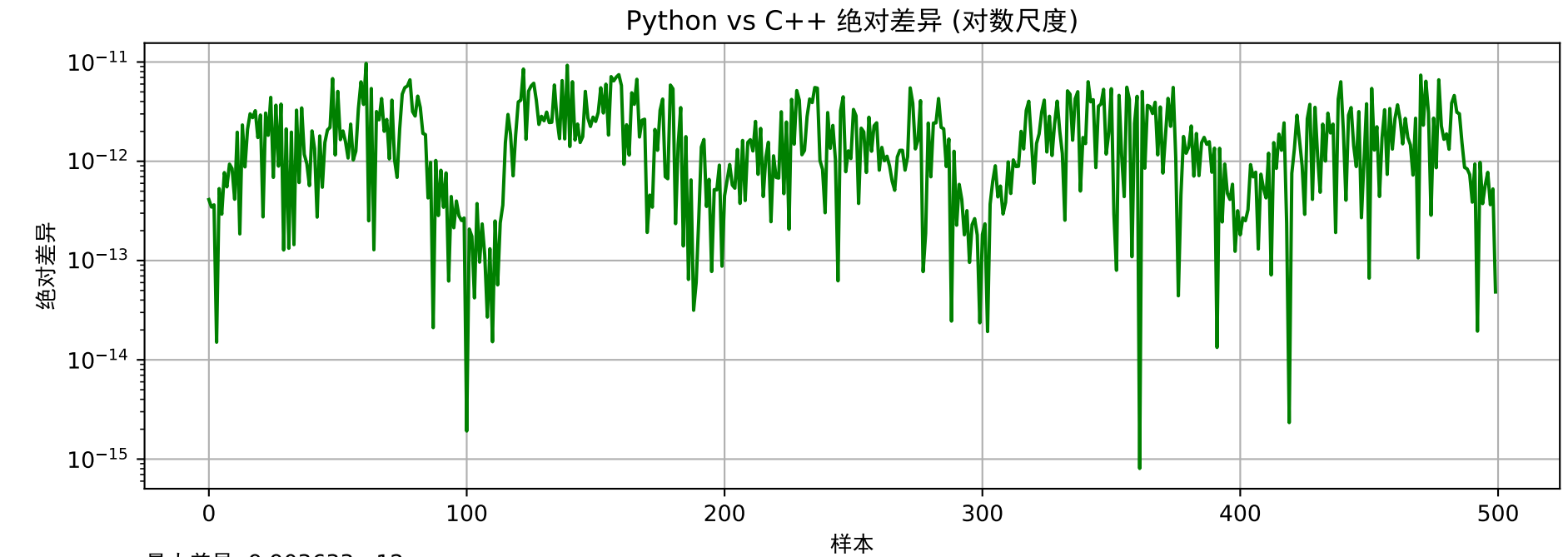
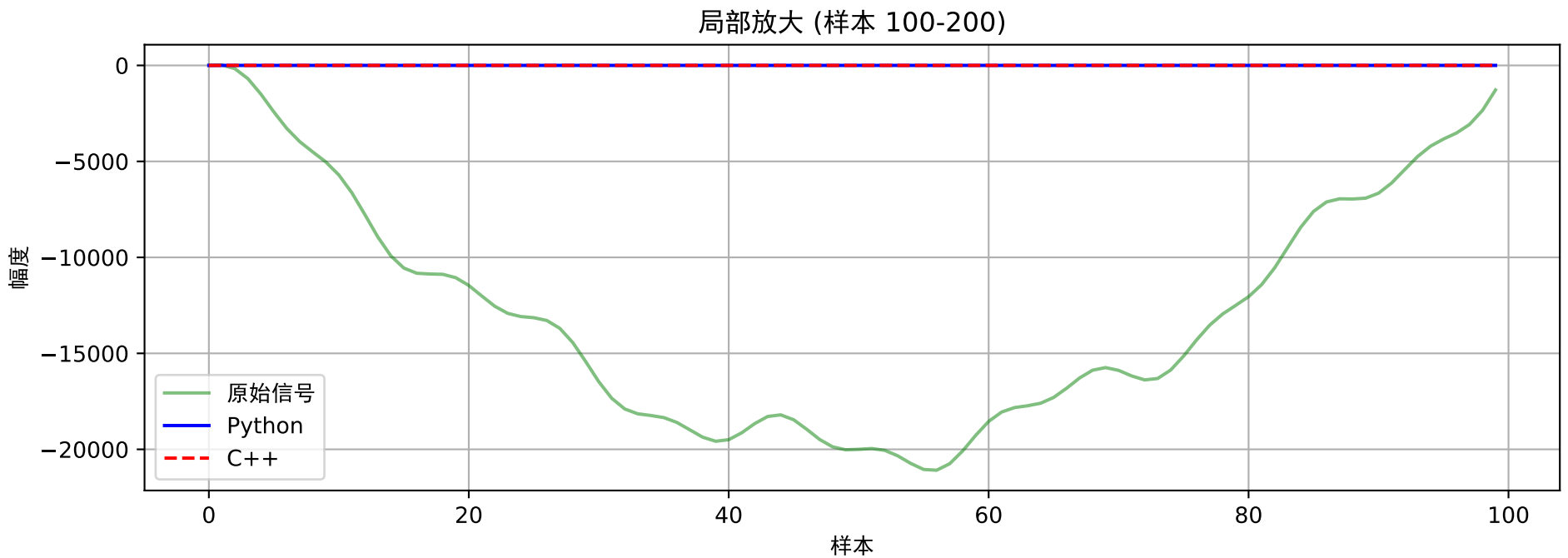
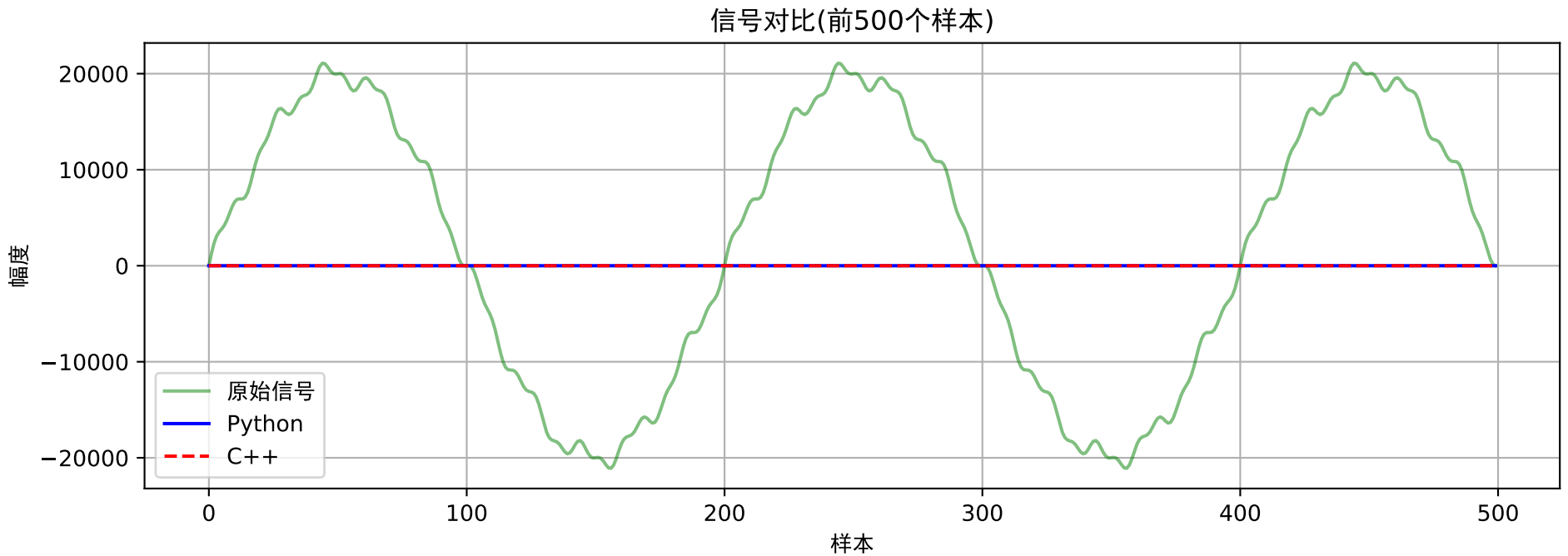
## Python vs C++ 绝对差异 (对数尺度)



最大差异: 2.354739e-11  
平均差异: 3.318944e-12  
标准差: 3.301992e-12

结果匹配: ✓

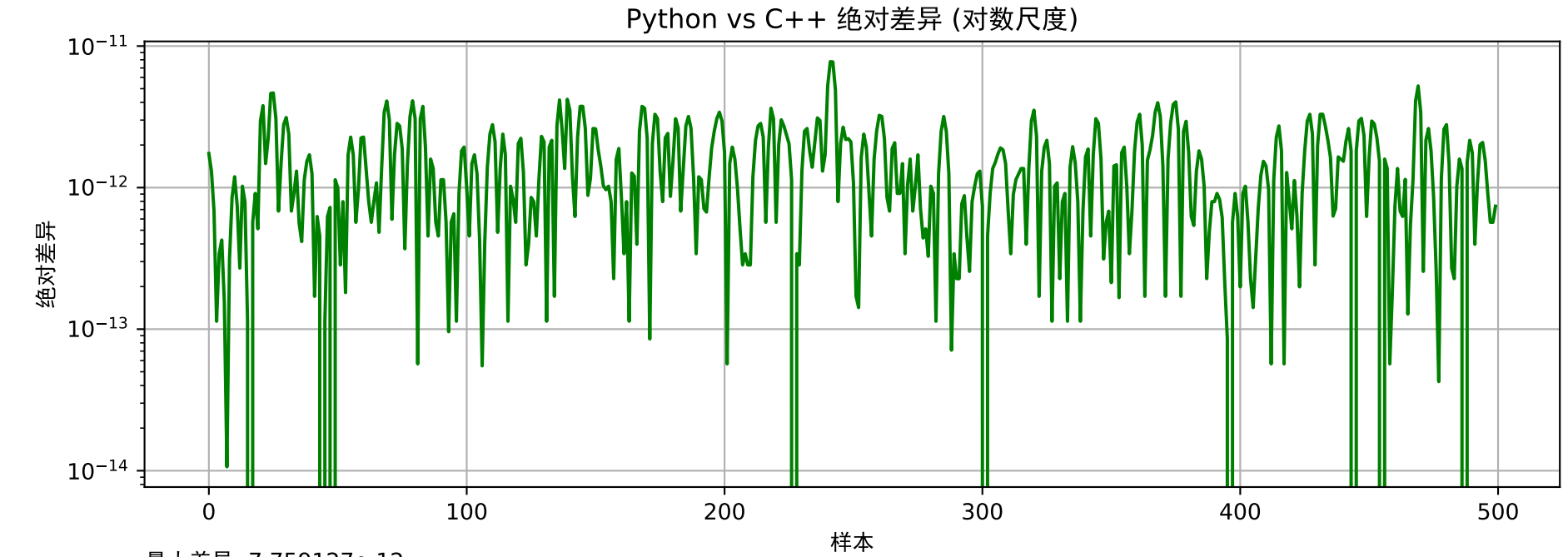
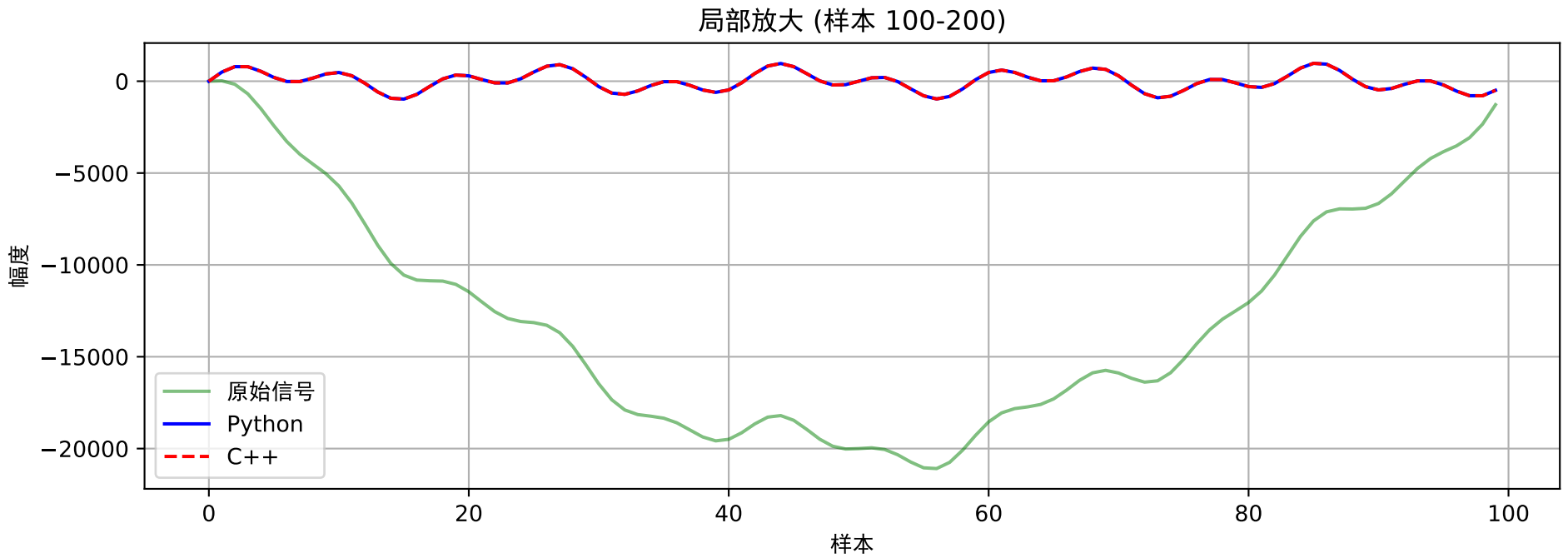
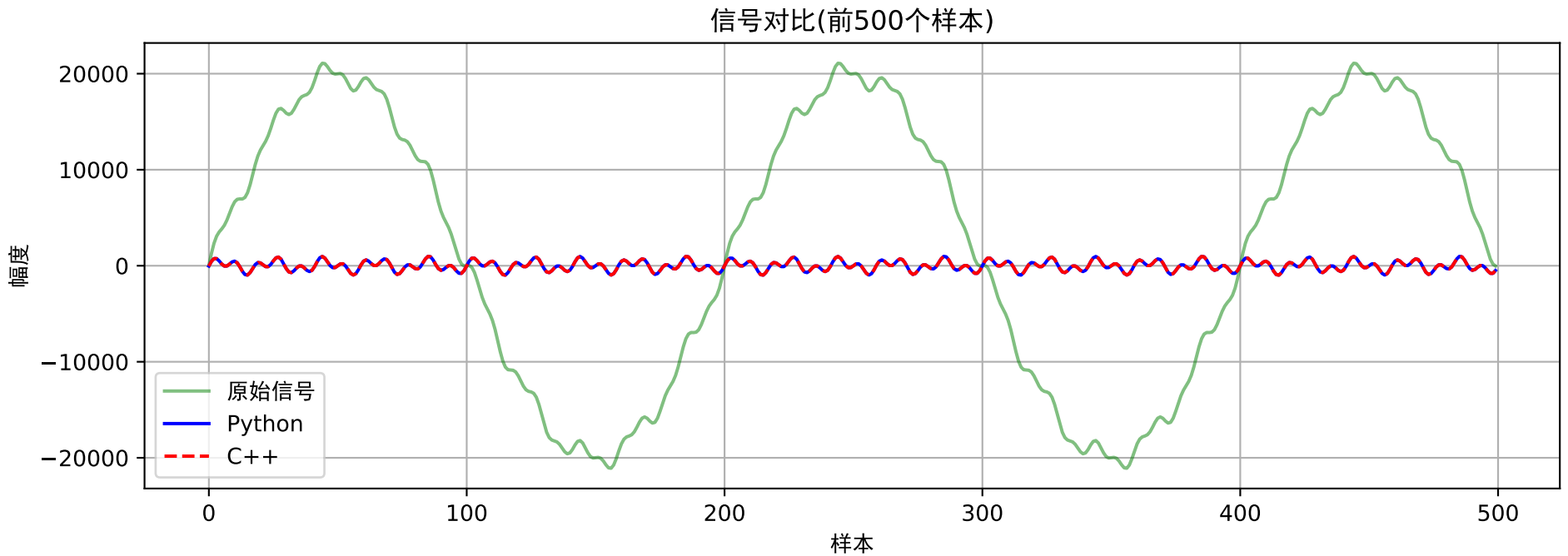
FiltFilt 滤波器实现比较: Python vs C++: 6th 阶高通滤波器 (Wn=0.4)



最大差异: 9.903633e-12  
平均差异: 1.974847e-12  
标准差: 1.914075e-12

结果匹配: ✓

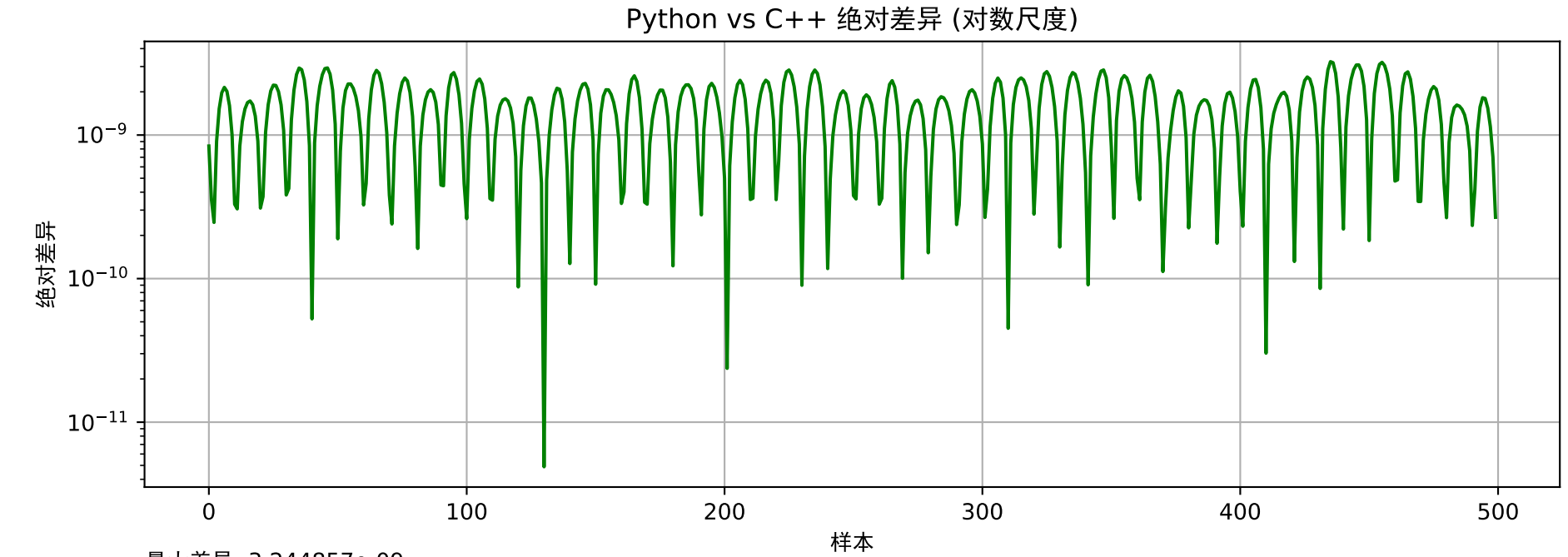
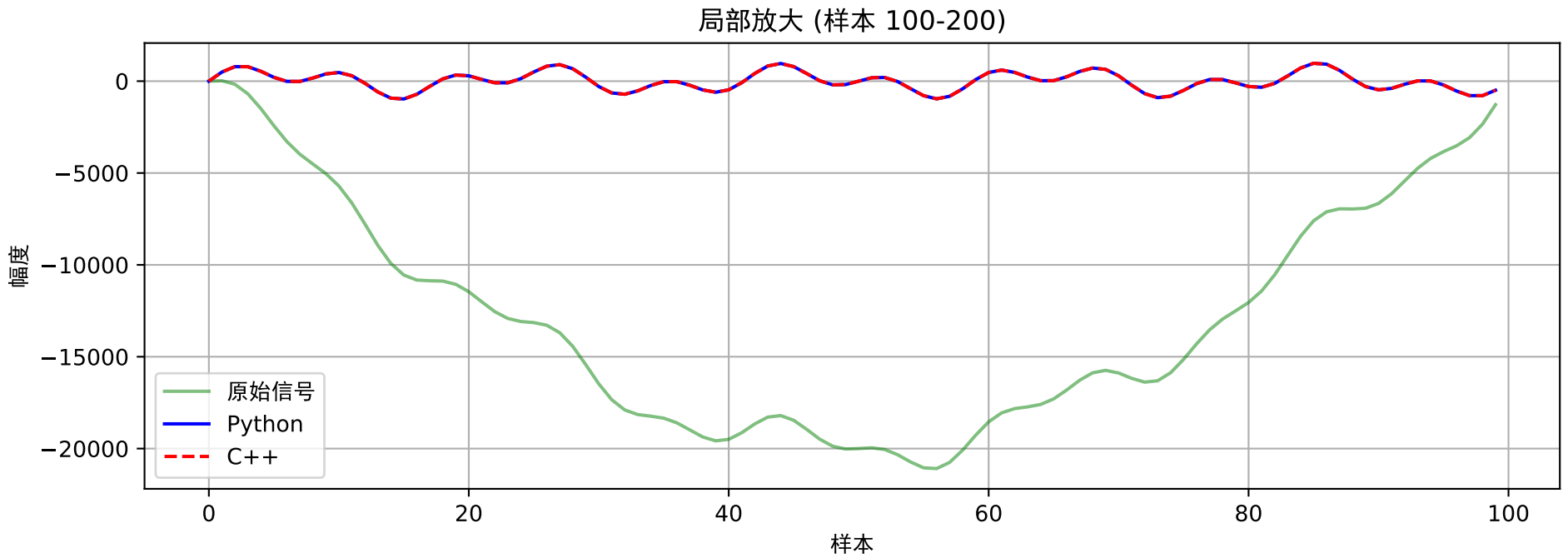
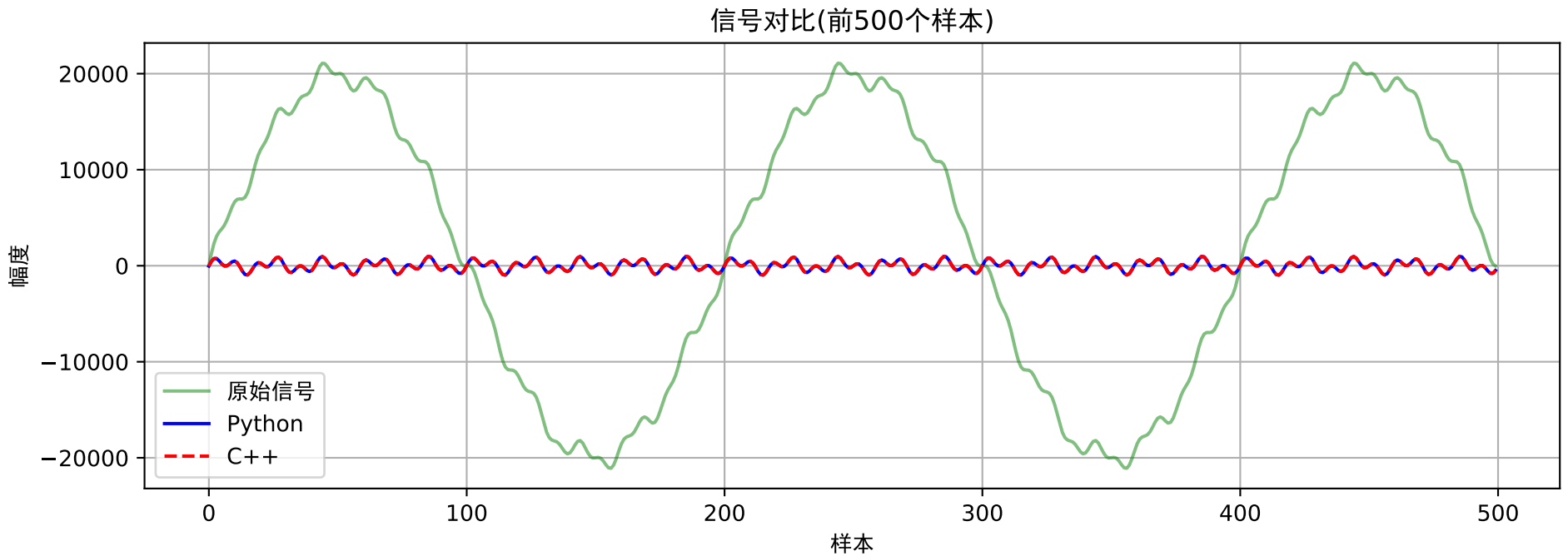
FiltFilt 滤波器实现比较: Python vs C++: 2nd 阶带通滤波器 (Wn=[0.1, 0.4])



最大差异: 7.759127e-12  
平均差异: 1.514996e-12  
标准差: 1.208998e-12

结果匹配: ✓

FiltFilt 滤波器实现比较: Python vs C++: 4th 阶带通滤波器 (Wn=[0.1, 0.3])

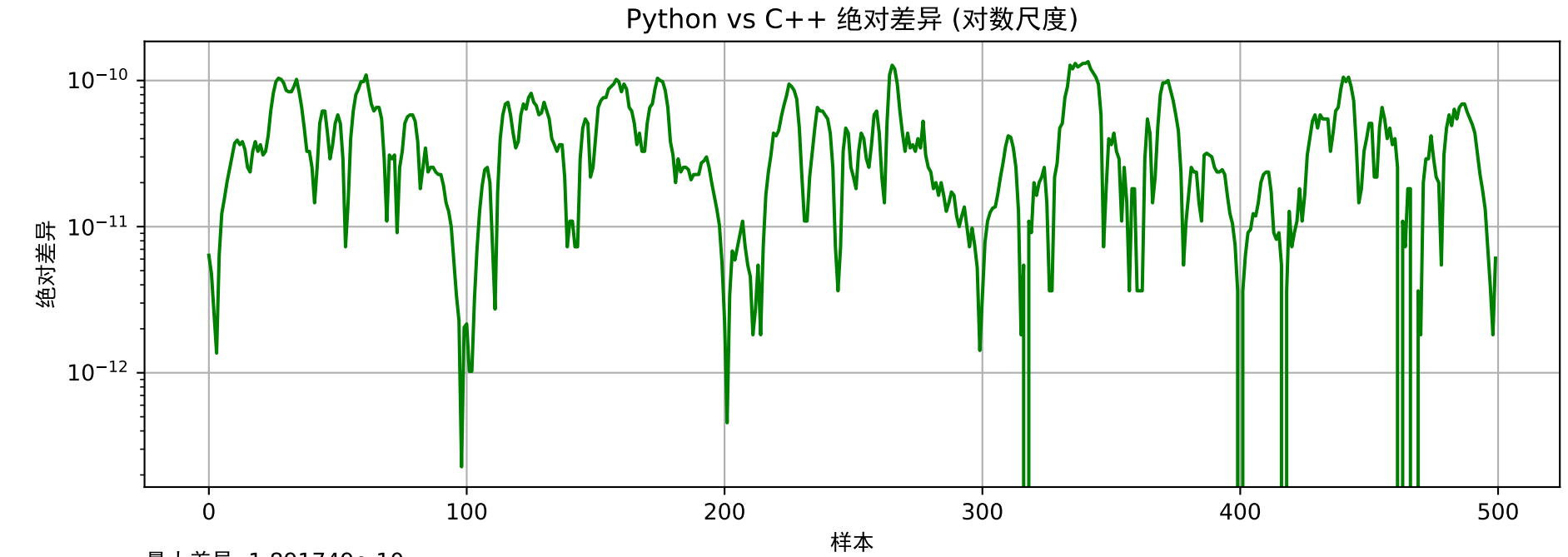
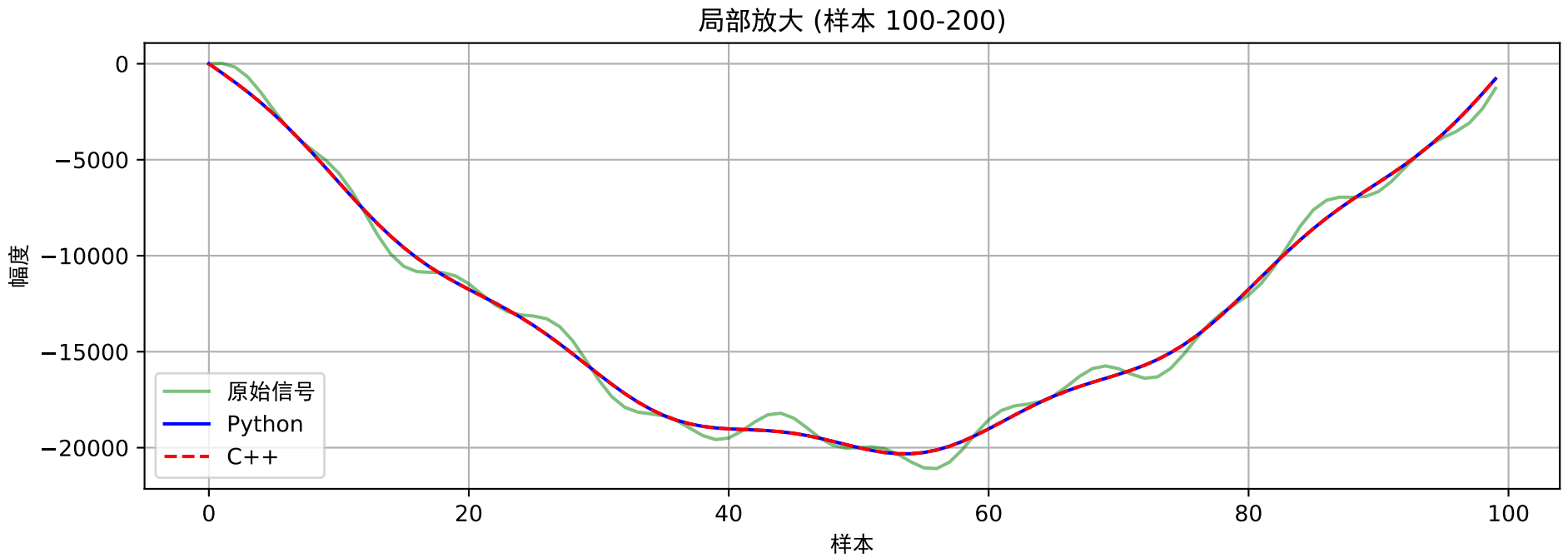
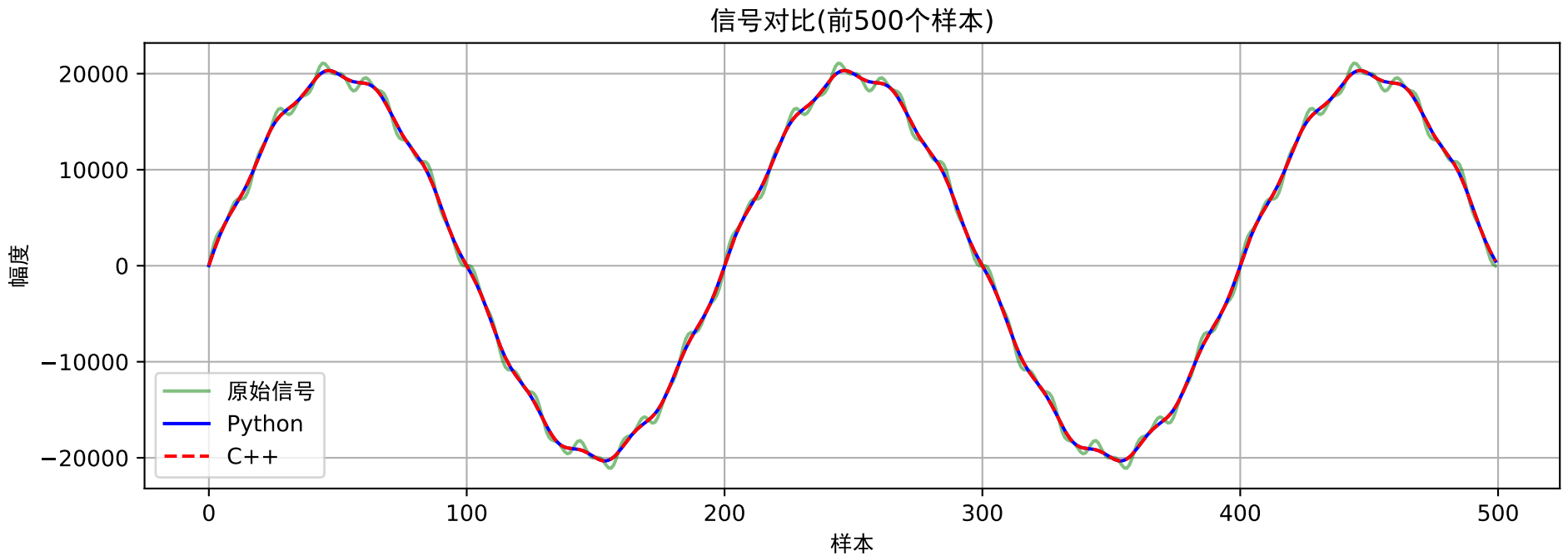


最大差异: 3.244857e-09  
平均差异: 1.383446e-09  
标准差: 7.400356e-10

结果匹配: ✗



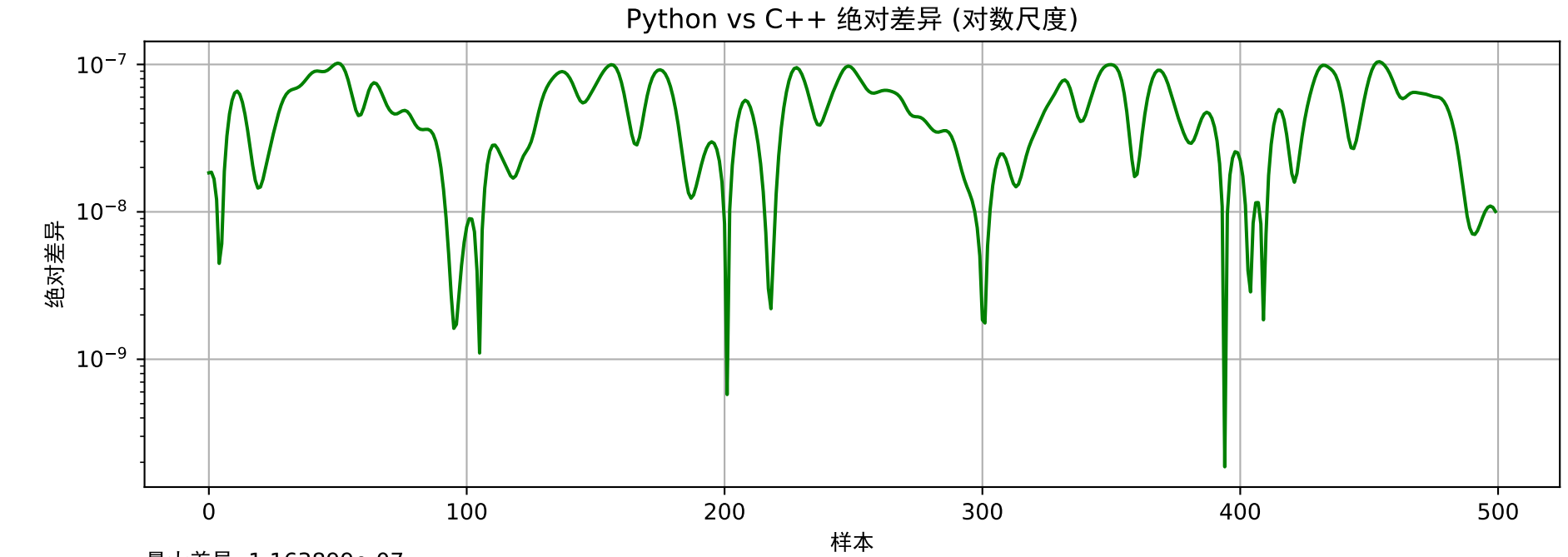
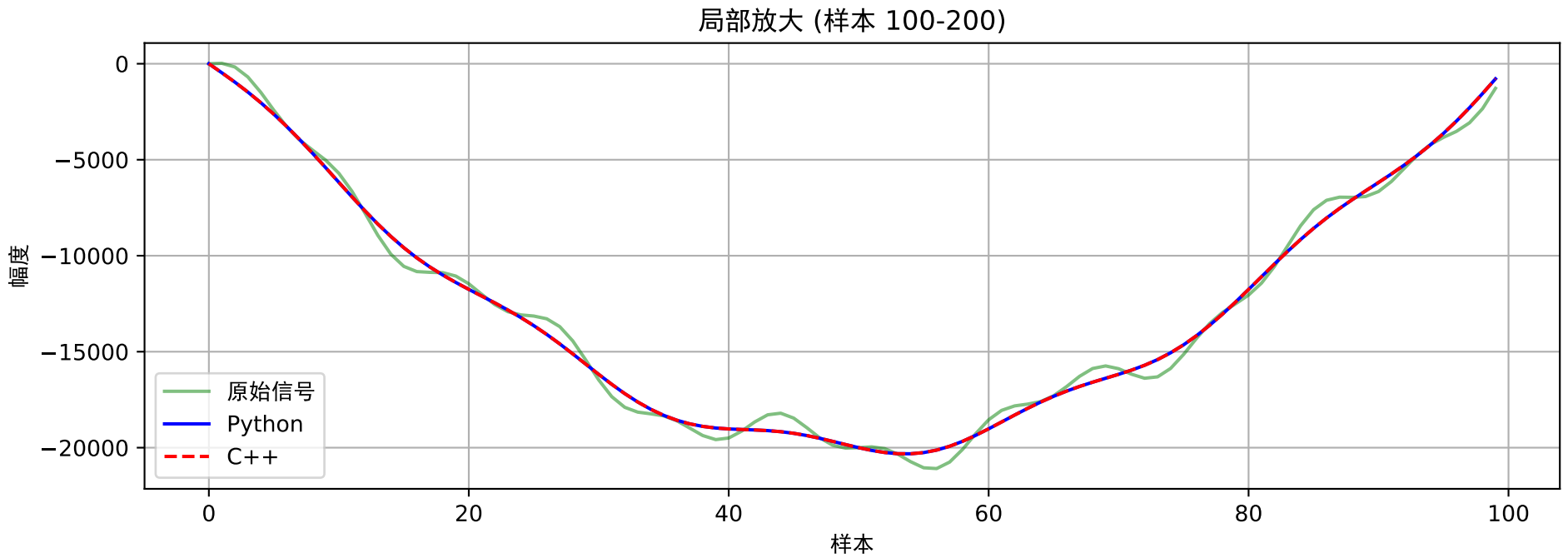
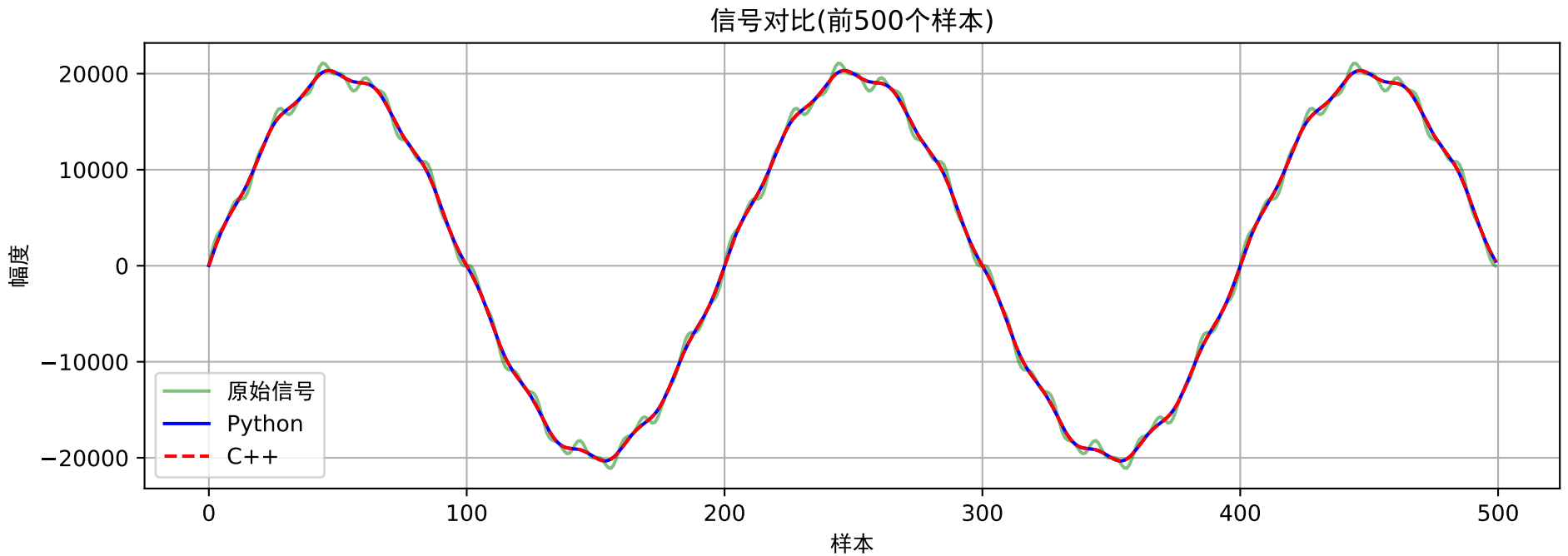
FiltFilt 滤波器实现比较: Python vs C++: 2nd 阶带阻滤波器 (Wn=[0.1, 0.4])



最大差异: 1.891749e-10  
平均差异: 3.868576e-11  
标准差: 3.282324e-11

结果匹配: ✗

FiltFilt 滤波器实现比较: Python vs C++: 4th 阶带阻滤波器 (Wn=[0.1, 0.3])



最大差异: 1.163899e-07  
平均差异: 4.651473e-08  
标准差: 2.732794e-08

结果匹配: ✗

## FiltFilt 滤波器实现比较: Python vs C++

|                                       |                |
|---------------------------------------|----------------|
| ✓ Test 1: 2nd 阶低通滤波器 (Wn=0.1)         | 最大差异: 5.46e-11 |
| ✗ Test 2: 4th 阶低通滤波器 (Wn=0.2)         | 最大差异: 1.56e-10 |
| ✗ Test 3: 6th 阶低通滤波器 (Wn=0.01)        | 最大差异: 3.90e-02 |
| ✓ Test 4: 2nd 阶高通滤波器 (Wn=0.15)        | 最大差异: 9.21e-12 |
| ✓ Test 5: 4th 阶高通滤波器 (Wn=0.25)        | 最大差异: 2.35e-11 |
| ✓ Test 6: 6th 阶高通滤波器 (Wn=0.4)         | 最大差异: 9.90e-12 |
| ✓ Test 7: 2nd 阶带通滤波器 (Wn=[0.1, 0.4])  | 最大差异: 7.76e-12 |
| ✗ Test 8: 4th 阶带通滤波器 (Wn=[0.1, 0.3])  | 最大差异: 3.24e-09 |
| ✗ Test 9: 2nd 阶带阻滤波器 (Wn=[0.1, 0.4])  | 最大差异: 1.89e-10 |
| ✗ Test 10: 4th 阶带阻滤波器 (Wn=[0.1, 0.3]) | 最大差异: 1.16e-07 |

总结: **5/10** 测试通过